



ROBERT H. SMITH SCHOOL OF BUSINESS

BUDT704: DATA PROCESSING AND ANALYSIS IN PYTHON

“Loan Approval Classification Dataset”

DATA SCIENCE PROJECT

By Group: 506-H

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INTRODUCTION

Background:

Predicting loan defaults is crucial for financial institutions to manage risk effectively and ensure sustainable lending practices. Loan defaults can lead to significant financial losses, impacting a lender's profitability and stability. By identifying customers likely to default, institutions can take preemptive measures such as adjusting credit terms, offering tailored repayment plans, or strengthening collection strategies.

Accurate predictions also enable better credit risk assessment, ensuring that resources are allocated efficiently and responsibly. This not only protects lenders but also promotes financial inclusion by providing fair access to credit for low-risk individuals. Furthermore, it contributes to overall economic stability by reducing the systemic risks associated with high default rates in financial markets.

Objective

The primary goal of the project is to predict whether a loan application will be approved or not based on various applicant and loan-specific features. The steps taken to achieve this goal are-

- **Understand Key Drivers:** Identify which factors (e.g., income, credit history, loan amount) most influence loan approval decisions.
- **Develop Classification Models:** Train and compare the accuracy of different machine learning algorithms to make predictions. The models created will be Linear Regression, Logistic regression, Decision trees and Random forest.
- **Solve Business Problems:** Mimic scenarios faced by financial institutions to automate decision-making processes or identify risky applicants.
- **Gain insights :** Based on the data analysis, gain insights into the customer demographic who might get rejected for loan applications.

DATASET DESCRIPTION:

The dataset contains 45000 rows and 14 variables.

1. **person_age:** Age of the person (in years).
2. **person_gender:** Gender of the person (e.g., male, female).
3. **person_education:** Highest education level attained by the person.
4. **person_income:** Annual income of the person (in dollars).
5. **person_emp_exp:** Total years of employment experience.
6. **person_home_ownership:** Person's home ownership status (e.g., rent, own, mortgage).
7. **loan_amnt:** Amount of the loan requested (in dollars).
8. **loan_intent:** Purpose or intent of the loan (e.g., education, medical, business).
9. **loan_int_rate:** Interest rate applied to the loan (as a percentage).

10. **loan_percent_income**: Loan amount as a percentage of the annual income.
11. **cb_person_cred_hist_length**: Length of the person's credit history (in years).
12. **credit_score**: Credit score of the person (numerical value).
13. **previous_loan_defaults_on_file**: Whether the person has defaulted on previous loans (yes/no).
14. **loan_status**: Target variable indicating loan approval (1 = approved, 0 = rejected).

DATA PROCESSING

1. DATA PROCESSING TASKS

1.1. Data Retrieval

The dataset, `loan_data.csv`, was loaded into the Python environment using the pandas library. The dataset contains 45,000 entries and 14 features, including demographic, financial, and credit-related attributes.

1.2. Data Cleaning

The dataset was cleaned to ensure quality and consistency:

- Data Type Validation and Conversion
 - Verified data types using the `info()` function to ensure appropriateness.
 - Converted `person_age` to integer format using `astype('int')`.
- Column Identification
 - Identified categorical and numerical columns for further processing.
- Age Categorization
 - Created `age_category` by binning `person_age` into categories (teen, adult, mature adult) using `pd.cut()`.
- Handling Anomalies
 - Removed records with `person_age > 100`.
 - Capped extreme outliers in `person_income` at the 99th percentile.
- Missing Values
 - Checked for missing values using `isnull().sum()`.
- Encoding Categorical Variables
 - Encoded `person_gender` (male: 1, female: 0) and `previous_loan_defaults_on_file` (Yes: 1, No: 0).
- Validation
 - Verified cleaned data using `head()`.

1.3. Exploratory Data Analysis

1.3.1. Summary Statistics

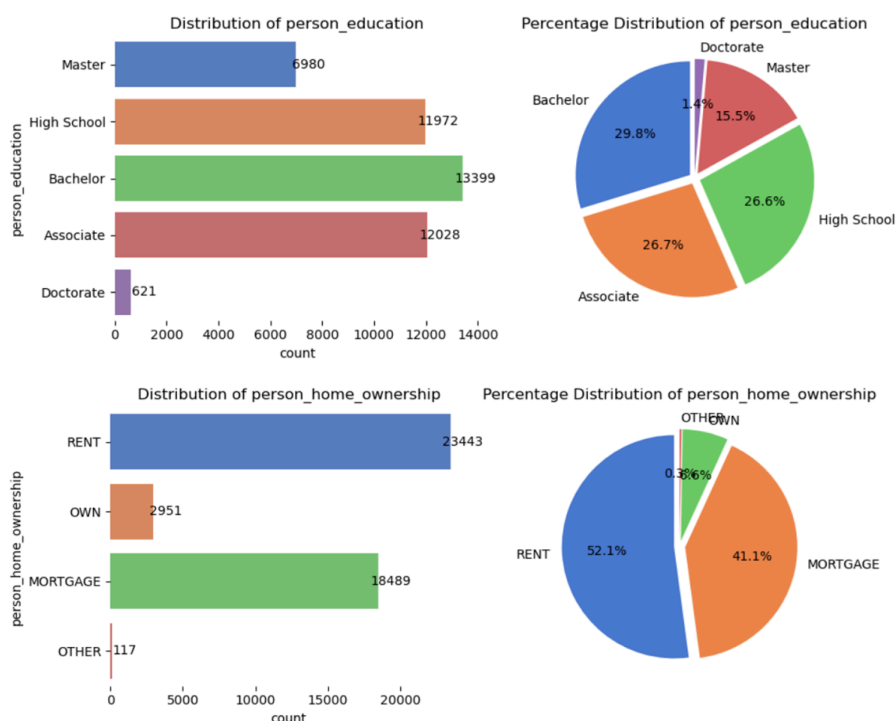
The `describe()` function of the Pandas library is employed to calculate and present summary statistics for numerical columns in the dataset. This gives us a broad understanding of how the

values vary in each of the fields. It also gives us a sense of how varied the data is.

	person_age	person_income	person_emp_exp	loan_amnt		loan_int_rate	loan_percent_income	cb_person_cred_hist_length
count	45000.000000	4.500000e+04	45000.000000	45000.000000	count	45000.000000	45000.000000	45000.000000
mean	27.764178	8.031905e+04	5.410333	9583.157556	mean	11.006606	0.139725	5.867489
std	6.045108	8.042250e+04	6.063532	6314.886691	std	2.978808	0.087212	3.879702
min	20.000000	8.000000e+03	0.000000	500.000000	min	5.420000	0.000000	2.000000
25%	24.000000	4.720400e+04	1.000000	5000.000000	25%	8.590000	0.070000	3.000000
50%	26.000000	6.704800e+04	4.000000	8000.000000	50%	11.010000	0.120000	4.000000
75%	30.000000	9.578925e+04	8.000000	12237.250000	75%	12.990000	0.190000	8.000000
max	144.000000	7.200766e+06	125.000000	35000.000000	max	20.000000	0.660000	30.000000

1.3.2 Data Visualization

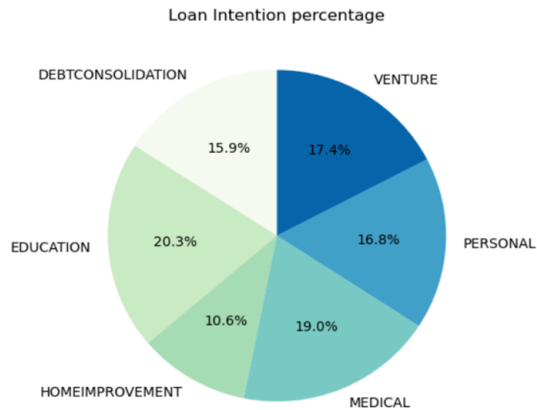
- Education and Home Ownership**



We observe that the majority of the customers have completed their education till the High-school, Bachelors or Associate level. Around 52% of the customers rent homes and 41% of them have taken out a mortgage.

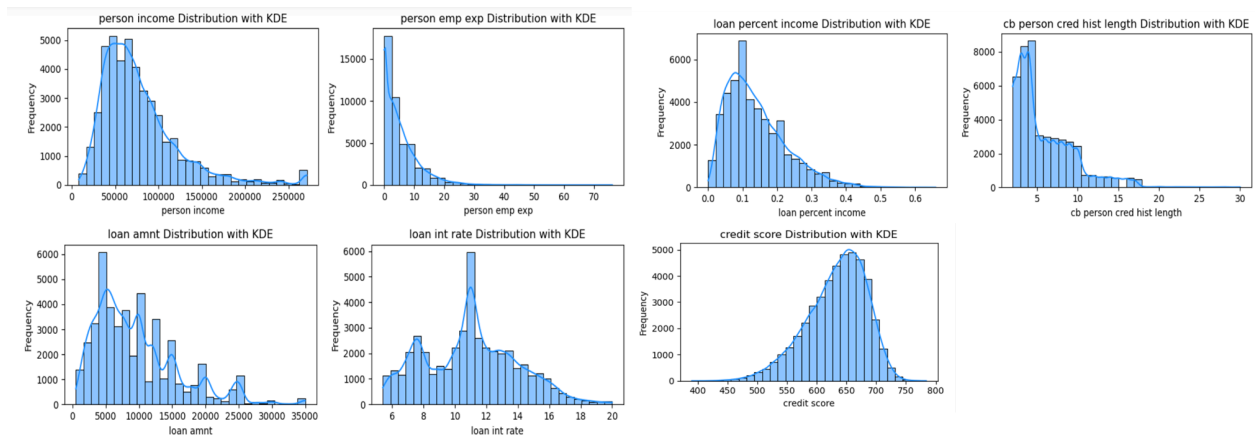
- Univariate Analysis - Loan Intent**

Based on the pie chart below, the types of loans taken out by the customers seem to be similar. Most customers take out education loans while home improvement loans are the least common among customers.

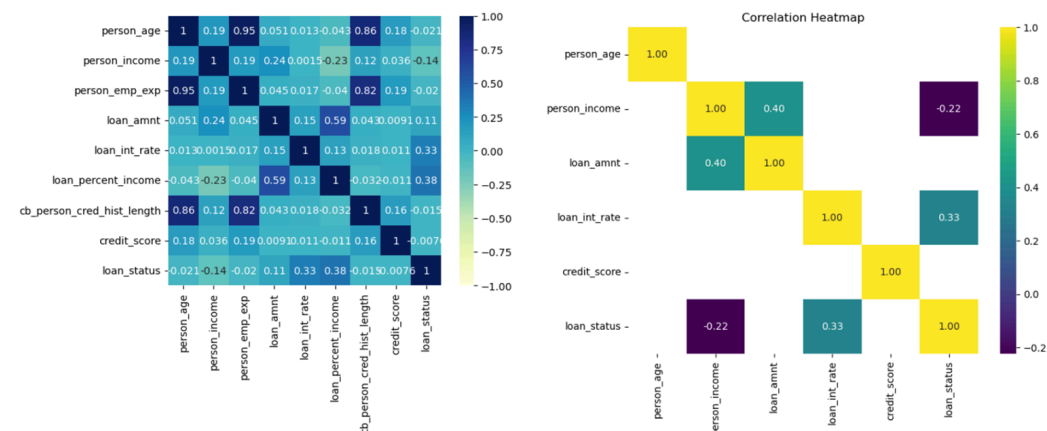


- Histogram plots of the numerical variables**

Most of the numerical variables follow a normal distribution, which is ideal. Some of the variables like credit history length and work experience are skewed which in the context of this problem is ideal as older people will have a higher credit history and more work experience than younger people.



- Correlation Heatmaps**



In the first heatmap, we find out the correlation between all the variables in the dataset. Based on this information, we narrow down and select the variables that are significantly (high or low) correlated than the others and create another heatmap. Strong positive correlation between age and employment experience (older borrowers have more work experience).

2. DATA ANALYSIS

2.1. Model Selection and Training

2.1.1 Linear Regression

Linear regression is a statistical technique used to explore and model the relationship between a dependent variable and one or more independent variables. In simple cases, it fits a straight line, while for multiple variables, it works with a hyperplane. The goal is to minimize the differences between predicted and actual values. However, since loan approval typically results in a binary outcome (approved or denied), linear regression is not usually the best choice for this type of task. It can still be useful for analyzing how variables interact, but it's not ideal for classification problems.

2.1.2 Logistic Regression

Logistic regression, unlike linear regression, is tailored for classification tasks. It's particularly useful for predicting binary outcomes—like whether a loan will be approved or not. By applying a logistic (or sigmoid) function, it transforms input variables into probabilities ranging from 0 to 1. This makes it well-suited for modeling loan approval, as it can estimate the likelihood of approval based on factors such as income, credit score, or requested loan amount. Logistic regression strikes a balance between simplicity, interpretability, and effectiveness, especially when the relationship between variables is approximately linear.

2.1.3 Decision Trees

Decision trees take a different approach. They use a rule-based system to split the dataset into subsets based on specific feature values, creating a tree-like structure. At each decision node, the algorithm selects the best split using metrics like the Gini index or entropy. The process continues until reaching leaf nodes that represent final outcomes, such as loan approval or denial. Decision trees are excellent for loan classification because they can handle non-linear relationships and provide clear, interpretable rules—making it easy to understand the reasoning behind a decision.

2.1.4 Random Forests

Random forests build on the strengths of decision trees by combining the predictions of multiple trees to enhance accuracy and minimize overfitting. Each tree is trained on a random sample of data and features, and the final output is determined by majority voting (for classification) or averaging (for regression). This ensemble method is highly effective for loan approval tasks. It handles complex and non-linear relationships, reduces variability through averaging, and is resistant to noise in the data, making it a robust and reliable choice.

2.2. **MODEL EVALUATION**

The models were evaluated using common metrics to assess their performance on the test dataset. Below are the evaluation steps and results:

2.2.1 **Data Splitting**

The dataset was split into training and testing sets (80% training, 20% testing) using the `train_test_split` function to ensure unbiased model evaluation.

2.2.2 **Evaluation Metrics**

Confusion Matrix

The confusion matrix was used to analyze the true positives, true negatives, false positives, and false negatives for each model.

Accuracy

Accuracy, calculated using the `accuracy_score` function, measures the ratio of correctly predicted instances to the total instances.

Classification Report

The `classification_report` provided precision, recall, F1-score, and support for each class:

- Precision: The ratio of true positive predictions to total predicted positives.
- Recall: The ratio of true positive predictions to total actual positives.
- F1-Score: The harmonic mean of precision and recall, balancing the two metrics.

2.2.3 **Model Evaluation Results**

Class 0 - represents 'Loan not approved'

Class 1 - represents 'Loan approved'

Model	Accuracy Score	Precision	Recall	F1-Score
Logistic Regression	0.825647	Class 0: 0.85 Class 1: 0.69	Class 0: 0.95 Class 1: 0.40	Class 0: 0.89 Class 1: 0.50
Decision Tree	0.900989	Class 0: 0.94 Class 1: 0.78	Class 0: 0.94 Class 1: 0.80	Class 0: 0.94 Class 1: 0.79
Random Forest	0.928770	Class 0: 0.94 Class 1: 0.90	Class 0: 0.98 Class 1: 0.77	Class 0: 0.96 Class 1: 0.83
Linear Regression	0.895322	N/A	N/A	N/A

Logistic Regression:

- The accuracy is relatively lower at 0.83, with a significant imbalance in performance between Class 0 and Class 1. Precision and recall for Class 0 are strong, but the model struggles with Class 1, showing a low recall (0.40) and precision (0.69). This means that while the model correctly identifies most of the Class 0 instances, it misses a significant portion of Class 1 instances, leading to many false negatives. Logistic Regression is not the best fit for this dataset, especially for predicting Class 1.

Decision Tree:

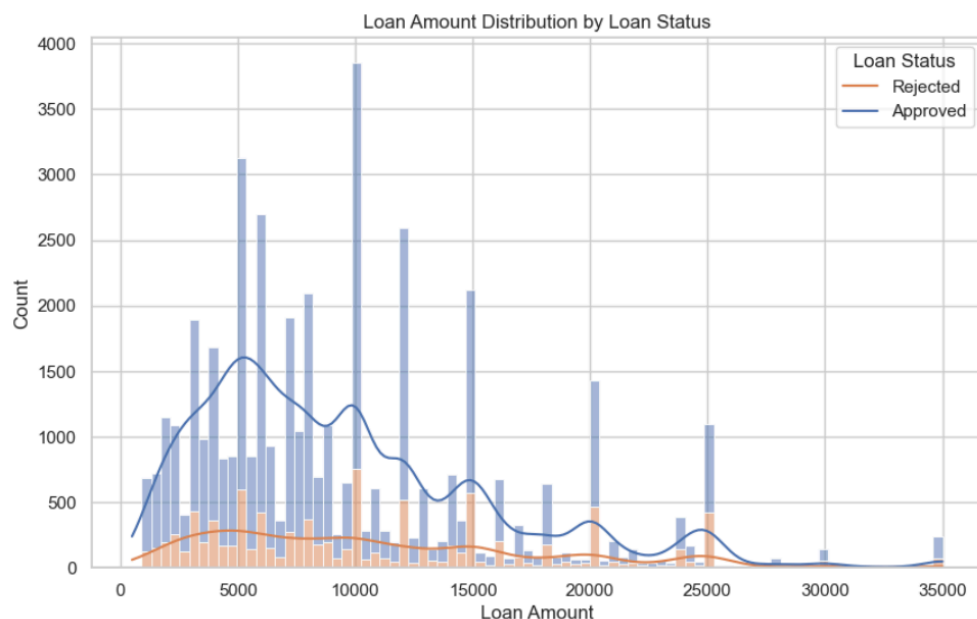
- The Decision Tree model shows high accuracy (0.90) and a well-balanced performance across both classes, with precision and recall values close to each other for both Class 0 and Class 1. This indicates that the Decision Tree is good at distinguishing between the two classes and performs well without much bias toward one class over the other.

Random Forest:

- Random Forest outperforms all other models with the highest accuracy (0.93). It demonstrates strong precision and recall for both classes, with Class 0 having excellent precision (0.98) and recall (0.94). Class 1 also shows strong performance, with precision of 0.90 and recall of 0.77. This makes Random Forest the most robust model for this dataset, as it performs well in identifying both Class 0 and Class 1 instances accurately.

BUSINESS TRANSACTIONS

1. What are the business opportunities in lending, targeting customers effectively, and improving loan approval processes?



Approval and Rejection Trends: Smaller loans (<\$10,000) have higher approval rates. As loan amounts increase, approvals decrease and rejections rise.

Peak Loan Amounts: Most approved loans are between \$5,000 and \$10,000, with a peak around \$10,000. Rejected loans are more spread out, with higher rejections between \$10,000–\$20,000.

Loan Amount Threshold: Approved loans sharply decrease beyond ~\$15,000, while rejections remain consistent.

Overall Distribution: Approved loans are more common in lower amounts; rejections increase with larger loans.

Conclusion: Smaller loans are more likely to be approved, while larger loans face higher rejection rates, reflecting cautious lending behavior as loan sizes grow.

2. Which features (e.g., interest rate, income, credit score) are most influential in predicting loan approval using machine learning models like logistic regression or decision trees?

Distributions: Most borrowers are in the 20-40 age range with moderate income (<\$100,000).

Loans are mainly between \$5,000–\$15,000 with typical interest rates around 10–12%.

Borrowers generally have fair-to-good credit scores (~600–700).

Target Market: Focus on younger borrowers with moderate incomes and fair credit scores.

Offer loans between \$5,000–\$15,000.

Risk Management: Safeguard approvals for higher credit score borrowers with lower loan-to-income ratios. Apply stricter checks for larger loans.

Customized Products: Offer low-interest loans for high-credit-score borrowers and short-term loans for borrowers with limited credit history.

Approval Strategies: Improve approval rates by targeting borrowers with credit scores above 600 and loan-to-income ratios under 30%.

Marketing: Tailor campaigns for first-time borrowers and highlight benefits like flexible repayment and lower interest rates for fair-to-good credit scores.

This analysis helps refine lending strategies, optimize loan products, and align marketing with borrower profiles.

RESULTS AND CONCLUSIONS

Key statistics:

This dataset has a high number of middle aged (around the 25-32 yrs old) customers, who fall in the middle-income bracket (average of 80000\$).

On average, loan amounts disbursed are approximately 5000\$, and credit scores are centered around 632.

Education loans contribute to the highest percentage of all loans, whilst home improvement loans contribute the least. Additionally, we can infer that these education loans are towards funding undergraduate tuition, which is seen by the age demographic

After preprocessing and modeling the dataset using various machine learning algorithms, the following insights and outcomes were observed:

Model Performance:

Four models were evaluated: Linear Regression, Logistic Regression, Decision Tree Classifier, and Random Forest Classifier. Key metrics such as accuracy, precision, and recall were computed to assess the models.

The Random Forest Classifier outperformed the others, showcasing the highest accuracy.

Feature Importance:

The Random Forest model identified the most significant predictors of loan approval:

Credit Score: Higher scores strongly correlate with loan approvals.

Loan Amount as a Percentage of Income: Lower percentages increase approval likelihood.

Length of Credit History: Longer histories contribute to higher approval rates.

Dataset Insights:

Applicants with higher incomes, lower loan-to-income ratios, and no history of default had a significantly higher chance of loan approval.

Education level and homeownership status also showed a minor influence on loan approval decisions.

Balanced Evaluation:

The Random Forest Classifier demonstrated robust performance across balanced datasets, maintaining high precision and recall across both approved and denied classes.

CONCLUDING STATEMENTS

Summing up our project, we explored loan approval predictions and identified crucial factors shaping financial decision-making processes. Our analysis highlighted the significant role of demographic, financial, and credit-related metrics in determining loan approval outcomes. Key insights revealed the importance of credit score, loan-to-income ratio, and credit history length in assessing applicant reliability and risk.

Among the machine learning models evaluated, the Random Forest Classifier stood out for its superior accuracy and ability to interpret feature importance, making it a strong candidate for practical applications in loan prediction systems. However, we recognize that real-world complexities, such as economic fluctuations or unobservable applicant traits, may require ongoing refinement of the predictive models.

This project underscores the potential of machine learning to enhance decision-making in financial services by improving the efficiency and accuracy of loan approval processes. Future work may include the integration of additional data sources and more sophisticated algorithms to further optimize predictive capabilities and address real-world challenges.

REFERENCE

<https://www.kaggle.com/datasets/taweilo/loan-approval-classification-data/code>